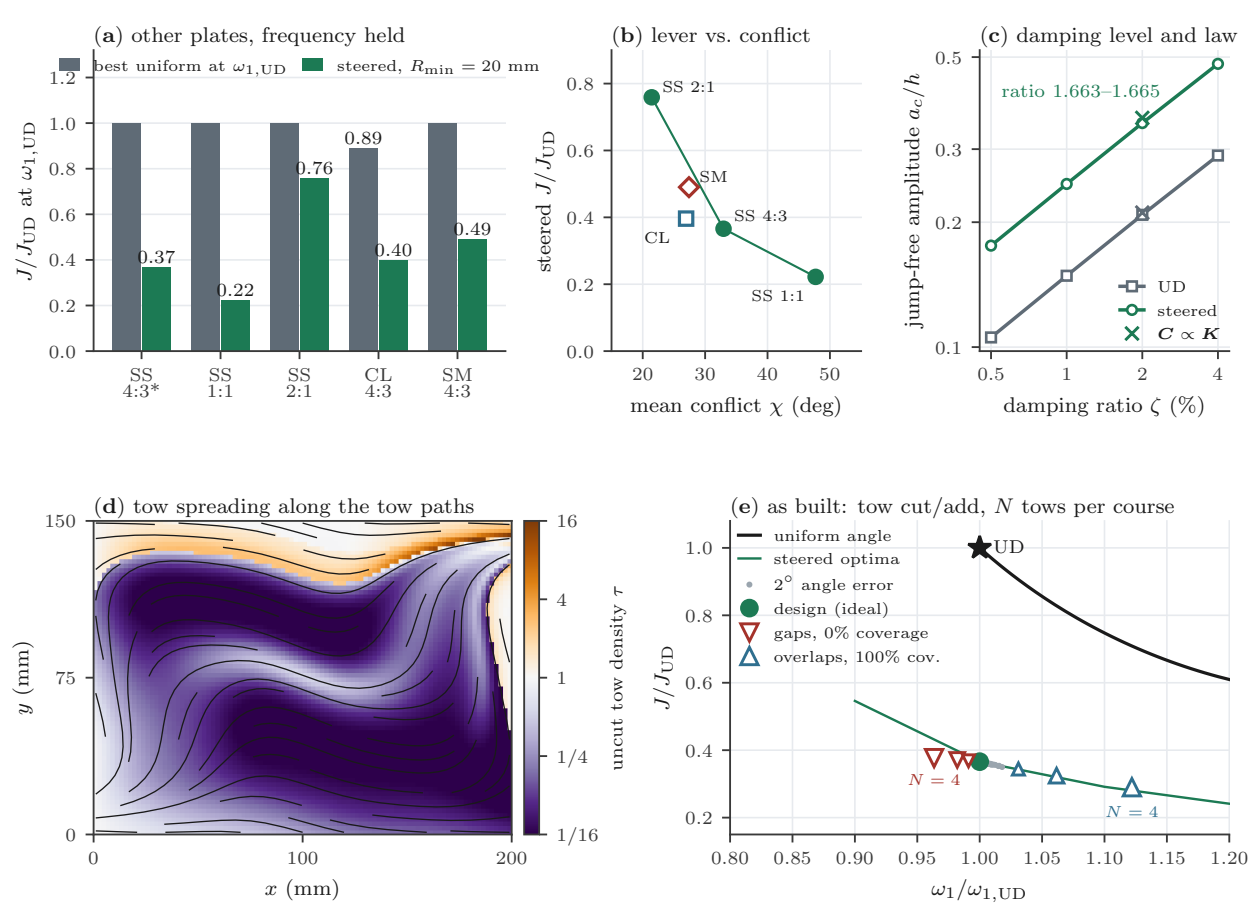

(a) other plates, frequency held
best uniform at $\omega_{1,\mathrm{UD}}$
steered, $R_{\min} = 20$ mm
J/J_{UD} at $\omega_{1,\mathrm{UD}}$
0.37
0.22
0.76
0.89
0.40
0.49
SS 4:3*
SS 1:1
SS 2:1
CL 4:3
SM 4:3
(b) lever vs. conflict
steered J/J_{UD}
SS 2:1
SM
CL
SS 4:3
SS 1:1
mean conflict χ (deg)
(c) damping level and law
jump-free amplitude a_c/h
ratio 1.663–1.665
UD
steered
$\boldsymbol{C} \propto \boldsymbol{K}$
damping ratio ζ (%)
(d) tow spreading along the tow paths
y (mm)
x (mm)
uncut tow density τ
(e) as built: tow cut/add, N tows per course
uniform angle
steered optima
2° angle error
design (ideal)
gaps, 0% coverage
overlaps, 100% cov.
UD
$N = 4$
$N = 4$
J/J_{UD}
$\omega_1/\omega_{1,\mathrm{UD}}$